INTELLIGENT ATTENDANCE CONTROL AND STUDENT PROGRESS MONITORING PLATFORM – EASWARI ERP PORTAL

PROJECT REPORT

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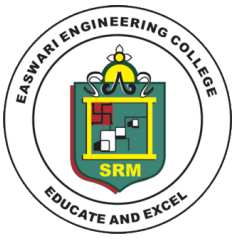
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BONAFIDE CERTIFICATE

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PROJECT REPORT

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ABSTRACT

In most colleges, attendance and internal marks are still kept in paper registers and spreadsheets that different people update at different times. The figures seen by a student, a class coordinator and the examination cell therefore often disagree, a class can be marked by a teacher who does not handle it, and short attendance is discovered too late to be corrected. This project presents the Easwari ERP Portal, a web platform for attendance control and student progress monitoring that keeps one authoritative record of each fact and controls who may change it. Student, Faculty and Administrator portals sit behind a single sign-in. New accounts receive a random temporary password that must be changed at the first login, and students may create a six-digit quick code that is locked after five wrong attempts.

Attendance is stored once for each student, subject offering, date and period under a database constraint that makes double marking impossible; the period, day and subject views are all derived from that record. The class coordinator may mark and correct every period of the class, a subject teacher may change only his or her own subject, and the administrator oversees every class, with the server enforcing these rules for every request. Internal marks are visible to students only after publication. Students see attendance, 75% eligibility, marks, practicals, university results, fees, leave, placement matches and a Career Hub. A separate FastAPI service computes an explainable academic-risk score.

The system uses Java 17 and Spring Boot 3.2.5, React 18 with Vite, JSON Web Tokens with BCrypt, an embedded H2 database that moves to PostgreSQL by configuration, and a Python scoring service. A trial institution of five departments, ten classes, 59 students and 25 staff members was seeded and exercised. In 68 recorded test cases and 11 unit tests the rules held in all but one case; the single defect found is reported with its fix.

KEYWORDS: Attendance Management, Student Progress Monitoring, College ERP, Role-Based Access Control, Spring Boot, React, JSON Web Token, Academic Risk Score, FastAPI

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GLOSSARY OF TECHNICAL TERMS

TERM	MEANING
API	Application Programming Interface
BCrypt	Blowfish-based password hashing function
CGPA	Cumulative Grade Point Average
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets
ERP	Enterprise Resource Planning
HOD	Head of the Department
HTTP	Hyper Text Transfer Protocol
IA	Internal Assessment
JPA	Java Persistence API
JSON	JavaScript Object Notation
JUnit	Java unit testing framework
JWT	JSON Web Token
OD	On Duty
ORM	Object Relational Mapping
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
SVG	Scalable Vector Graphics
UI	User Interface
URL	Uniform Resource Locator